

Binomial Coefficients

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COMP2711 Discrete Mathematical Tools

1 The binomial theorem

Recall that $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the k -element subsets of an n -element set, for integers $0 \leq k \leq n$.

Must understand: The binomial theorem. For every integer $n \geq 0$,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

To obtain $x^{n-k}y^k$, choose y from k of the n factors and x from the rest. There are $\binom{n}{k}$ choices, and each contributes the same term.

Example: expanding a cube. Multiplying one factor at a time gives

$$\begin{aligned}(x + y)^3 &= (x^2 + xy + yx + y^2)(x + y) \\ &= x^3 + x^2y + xyx + xy^2 + yx^2 + yxy + y^2x + y^3 \\ &= x^3 + 3x^2y + 3xy^2 + y^3.\end{aligned}$$

We can find the coefficient of xy^2 without listing all eight products. Choose the two factors supplying y :

Factor 1	Factor 2	Factor 3	→	Product
x	y	y	→	xy^2
y	x	y	→	xy^2
y	y	x	→	xy^2

There are $\binom{3}{2} = 3$ choices, so the coefficient is 3.

Example: expanding $(1 + x)^4$. Choosing x from k factors gives $\binom{4}{k}x^k$, so

$$(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4.$$

Similarly, the coefficient of x^3 in $(1 + x)^5$ is $\binom{5}{3} = 10$: choose x from three factors and 1 from the other two.

2 The generalized form

For constants a, b , replacing x, y by ax, by gives, for every integer $n \geq 0$,

$$(ax + by)^n = \sum_{k=0}^n \binom{n}{k} (ax)^{n-k} (by)^k = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k x^{n-k} y^k.$$

The coefficient of $x^{n-k}y^k$ is $\binom{n}{k}a^{n-k}b^k$: the factor choices contribute $\binom{n}{k}$, and the chosen terms contribute the numerical powers.

Must understand: Coefficient extraction. Write the general term, match its exponent to the requested exponent, then evaluate its coefficient. Keep all numerical powers and signs.

Example: the coefficient of x^5 in $(2x - 1)^8$. If k factors supply -1 , the other $8 - k$ factors supply $2x$. The general term is

$$\binom{8}{k} (2x)^{8-k} (-1)^k = \binom{8}{k} 2^{8-k} (-1)^k x^{8-k}.$$

Set $8 - k = 5$, giving $k = 3$. The coefficient is

$$\binom{8}{3} 2^5 (-1)^3 = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} \cdot 32 \cdot (-1) = -1792.$$

Example: the constant term in $(x^2 + x^{-1})^9$, where $x \neq 0$. The binomial theorem also applies when its two terms are powers of x . Choosing x^{-1} from k factors gives

$$\binom{9}{k} (x^2)^{9-k} (x^{-1})^k = \binom{9}{k} x^{18-3k}.$$

A constant term has exponent zero. Set $18 - 3k = 0$, giving $k = 6$. The constant term is $\binom{9}{6} = 84$.

Follow-up: the coefficient of x^4 . The equation $18 - 3k = 4$ gives $k = 14/3$, which is not an integer. No choice of factors produces x^4 , so its coefficient is 0.

The index must be an integer between 0 and n . If no such index gives the requested exponent, the term is absent.

Practice. Find the coefficient of x^2 in $(1 + 3x)^4$. Choose $3x$ twice: the coefficient is $\binom{4}{2} 3^2 = 54$.

3 Four views of selection

Must understand: Combinatorial proof. Prove an identity by counting the same collection of outcomes in two ways. Explain what each expression counts and why each outcome is counted exactly once.

Our running example is a four-person team from $\{A, B, C, D, E, F\}$. A team is an unordered subset. Count the same teams directly, then split them into cases in three ways.

View	Count of k -element subsets
Direct selection	$\binom{n}{k}$
Include or exclude one element	$\binom{n-1}{k-1} + \binom{n-1}{k}$
Largest selected label	$\sum_{j=k}^n \binom{j-1}{k-1}$
Selection from two groups	$\sum_{i=0}^k \binom{a}{i} \binom{b}{k-i} \quad (a+b=n)$

Here $1 \leq k < n$ and a, b are nonnegative integers with $a+b=n$. An impossible choice has count 0: $\binom{u}{v} = 0$ if $v < 0$ or $v > u$. The identities below state their individual ranges.

3.1 Direct selection

Choose four of the six people:

$$\binom{6}{4} = \frac{6!}{4!2!} = 15.$$

The next three views give the same answer because they count these same teams.

3.2 Include or exclude: Pascal's identity

Split the teams by whether A is selected:

Case	Remaining choice	Count
Include A	Choose three from B, C, D, E, F	$\binom{5}{3} = 10$
Exclude A	Choose four from B, C, D, E, F	$\binom{5}{4} = 5$

Every team belongs to exactly one case, so

$$\binom{6}{4} = \binom{5}{3} + \binom{5}{4} = 10 + 5 = 15.$$

Pascal's identity. For integers $n \geq 2$ and $1 \leq k < n$,

$$\underbrace{\binom{n}{k}}_{\text{all selections}} = \underbrace{\binom{n-1}{k-1}}_{\text{include a fixed element}} + \underbrace{\binom{n-1}{k}}_{\text{exclude that element}}.$$

Including the fixed element leaves $k - 1$ elements to choose from the remaining $n - 1$. Excluding it leaves k to choose. These disjoint cases cover every selection.

Pascal's triangle. Together with $\binom{n}{0} = \binom{n}{n} = 1$, Pascal's identity builds the following table. Row n contains the coefficients of $(x + y)^n$.

row 0	1								
row 1	1	1							
row 2	1	2	1						
row 3	1	3	3	1					
row 4	1	4	6	4	1				
row 5	1	5	10	10	5	1			
row 6	1	6	15	20	15	6	1		

$10 + 5 = 15$

In this right-triangle layout, each interior entry adds the entries immediately above it and above-left.

3.3 Largest label: the hockey-stick identity

Relabel A, B, C, D, E, F as $1, 2, 3, 4, 5, 6$. Each four-person team has exactly one largest label: 4, 5, or 6.

Largest label	Choose the other three from	Count
4	$\{1, 2, 3\}$	$\binom{3}{3} = 1$
5	$\{1, 2, 3, 4\}$	$\binom{4}{3} = 4$
6	$\{1, 2, 3, 4, 5\}$	$\binom{5}{3} = 10$

Thus

$$\binom{6}{4} = \binom{3}{3} + \binom{4}{3} + \binom{5}{3} = 1 + 4 + 10 = 15.$$

Hockey-stick identity. For integers $1 \leq k \leq n$,

$$\binom{n}{k} = \sum_{j=k}^n \binom{j-1}{k-1}.$$

Count k -element subsets of $\{1, \dots, n\}$ by their largest element j . Once j is fixed, choose the other $k - 1$ elements from the $j - 1$ smaller labels. Summing over $j = k, \dots, n$ counts every subset once.

The highlighted entries form a “hockey stick” in Pascal’s triangle. The same sum follows by repeatedly applying Pascal’s identity:

$$\begin{array}{ccccccc}
 1 & & & & & & \\
 1 & 1 & & & & & \\
 1 & 2 & 1 & & & & \\
 1 & 3 & 3 & 1 & & & \\
 1 & 4 & 6 & 4 & 1 & & \\
 1 & 5 & 10 & 10 & 5 & 1 & \\
 1 & 6 & 15 & 20 & 15 & 6 & 1
 \end{array}
 \qquad
 \begin{aligned}
 \binom{3}{3} &= \binom{4}{4} = 1, \\
 1 + 4 &= \binom{5}{4} = 5, \\
 1 + 4 + 10 &= \binom{6}{4} = 15.
 \end{aligned}$$

3.4 Two groups: Vandermonde’s identity

Keep the same four-person teams. Split the six people into disjoint groups $\{A, B, C\}$ and $\{D, E, F\}$, and count by how many members come from the first group.

From Group 1	From Group 2	Choices	Count
1	3	$\binom{3}{1}\binom{3}{3}$	3
2	2	$\binom{3}{2}\binom{3}{2}$	9
3	1	$\binom{3}{3}\binom{3}{1}$	3

Selecting 0 or 4 from Group 1 is impossible: one group would need to supply four people. Multiply within each case, then add:

$$\binom{6}{4} = \binom{3}{1}\binom{3}{3} + \binom{3}{2}\binom{3}{2} + \binom{3}{3}\binom{3}{1} = 3 + 9 + 3 = 15.$$

Vandermonde’s identity. For nonnegative integers a, b with $n = a + b$, and $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{a+b}{k} = \sum_{i=0}^k \binom{a}{i} \binom{b}{k-i}.$$

Here $\binom{u}{v} = 0$ when $v < 0$ or $v > u$. The left side chooses k people from two disjoint groups of sizes a, b . The term indexed by i chooses i from the first and $k - i$ from the second. Every team has exactly one such value of i .

Pascal as a special case. Put A alone in Group 1 and the other $n - 1$ people in Group 2. Only $i = 0, 1$ are possible:

$$\binom{n}{k} = \binom{1}{0}\binom{n-1}{k} + \binom{1}{1}\binom{n-1}{k-1} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Application: a sum of squares. Take two groups of n people and select n people in total. Since $\binom{n}{n-i} = \binom{n}{i}$, Vandermonde's identity gives

$$\binom{2n}{n} = \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i} = \sum_{i=0}^n \binom{n}{i}^2.$$

For $n = 3$, this reads $\binom{6}{3} = 1^2 + 3^2 + 3^2 + 1^2 = 20$. This application selects three people from six, unlike the four-person running example.

4 Corollaries of the binomial theorem

Substitute constants for x, y in

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

Each resulting identity also has a counting interpretation.

x	y	Result
1	1	$(1 + 1)^n = 2^n = \sum_{k=0}^n \binom{n}{k}$
1	-1	$(1 - 1)^n = 0 = \sum_{k=0}^n (-1)^k \binom{n}{k} \quad (n \geq 1)$
1	2	$(1 + 2)^n = 3^n = \sum_{k=0}^n 2^k \binom{n}{k}$

4.1 All subsets

How many subsets does $\{A, B, C\}$ have? Grouping by size gives counts 1, 3, 3, 1, totaling 8. Alternatively, each element is included or excluded, giving $2^3 = 8$.

The same two counts for an n -element set give

$$\sum_{k=0}^n \binom{n}{k} = 2^n \quad (n \geq 0).$$

The left side counts by subset size; the right side chooses membership for each element.

Algebraic proof. Set $x = y = 1$:

$$\underbrace{(1 + 1)^n}_{2^n} = \sum_{k=0}^n \binom{n}{k} \underbrace{1^{n-k} 1^k}_1 = \sum_{k=0}^n \binom{n}{k}.$$

4.2 Even-size and odd-size subsets

Does a nonempty finite set S have more even-size subsets or more odd-size subsets? Fix $A \in S$. For every even-size subset $T \subseteq S$, define

$$f(T) = \begin{cases} T \setminus \{A\}, & A \in T, \\ T \cup \{A\}, & A \notin T. \end{cases}$$

Adding or removing A changes the size by one, so $f(T)$ has odd size. Switching A 's membership again recovers T . Thus every odd-size subset has exactly one preimage, and f is a bijection. For $S = \{A, B, C\}$:

$$\begin{array}{ccc} \text{even size} & \longleftrightarrow & \text{odd size} \\ \emptyset & \longleftrightarrow & \{A\} \\ \{A, B\} & \longleftrightarrow & \{B\} \\ \{A, C\} & \longleftrightarrow & \{C\} \\ \{B, C\} & \longleftrightarrow & \{A, B, C\} \end{array}$$

For $|S| = n \geq 1$, the two counts are equal and sum to 2^n . Hence

$$\underbrace{\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots}_{\text{even sizes up to } n} = \underbrace{\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots}_{\text{odd sizes up to } n} = 2^{n-1}.$$

Subtract the odd-size count from the even-size count:

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \cdots = \sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

The sign $(-1)^k$ is $+1$ for even k and -1 for odd k .

Algebraic proof. Set $x = 1$, $y = -1$:

$$0 = (1 - 1)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} (-1)^k = \sum_{k=0}^n (-1)^k \binom{n}{k}.$$

The condition $n \geq 1$ matters: the empty set has one even-size subset and no odd-size subset.

4.3 Weighted sums

Example: two labels or no label. Each of three students gets a red label, a blue label, or no label. There are $3^3 = 27$ assignments. Alternatively, choose the k recipients first, then their colors:

$$\binom{3}{0} 2^0 + \binom{3}{1} 2^1 + \binom{3}{2} 2^2 + \binom{3}{3} 2^3 = 1 + 6 + 12 + 8 = 27.$$

For n students, this gives

$$\sum_{k=0}^n \binom{n}{k} 2^k = 3^n \quad (n \geq 0).$$

There are $\binom{n}{k}$ choices of recipients and 2^k choices of label colors. Summing over k counts every assignment once.

Algebraic proof. Set $x = 1$, $y = 2$:

$$3^n = (1 + 2)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} 2^k = \sum_{k=0}^n \binom{n}{k} 2^k.$$

Practice: three colors or no label. Each student now has four choices. Alternatively, choose k recipients and one of three colors for each:

$$4^n = \sum_{k=0}^n \binom{n}{k} 3^k.$$

5 Counting applications

5.1 A committee and its chair

Problem. From $n \geq 1$ people, choose a nonempty committee and designate one of its members as chair. How many outcomes are possible?

The committee may have any size from 1 to n . The same members with a different chair count as a different outcome.

Committee first. For each size k , choose the members in $\binom{n}{k}$ ways and the chair in k ways. The total is $\sum_{k=1}^n k \binom{n}{k}$.

Chair first. Choose the chair in n ways. Each other person may join or stay out, giving 2^{n-1} committees for that chair. The total is $n2^{n-1}$.

Both methods count each committee with its chair exactly once, so

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}.$$

For three people, the first count is $1 \cdot 3 + 2 \cdot 3 + 3 \cdot 1 = 12$; the second is $3 \cdot 2^2 = 12$.

5.2 Optional: complement pairs

Problem. How many subsets of $\{1, 2, 3, 4, 5\}$ have at most two elements? Directly, the count is $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} = 1 + 5 + 10 = 16$.

Alternatively, pair each subset with its complement. Their sizes add to five, so exactly one subset in each pair has size at most two. For example,

$$\begin{array}{lll} \emptyset & \longleftrightarrow & \{1, 2, 3, 4, 5\} \\ \{1\} & \longleftrightarrow & \{2, 3, 4, 5\} \\ \{1, 4\} & \longleftrightarrow & \{2, 3, 5\} \end{array}$$

Exactly half of the 2^5 subsets are counted, giving $2^5/2 = 16$.

General statement. For every integer $n \geq 1$,

$$\sum_{k=0}^{n-1} \binom{2n-1}{k} = 2^{2n-2}.$$

Proof. Let $|S| = 2n - 1$. Pair $T \subseteq S$ with $S \setminus T$. Their sizes add to $2n - 1$, so exactly one has size at most $n - 1$. Complementing twice returns T , and no subset is its own complement because S is nonempty. Thus the 2^{2n-1} subsets form 2^{2n-2} pairs, each contributing one desired subset.

This pairing compares small and large subsets. The even/odd proof instead changes the membership of one fixed element.

6 Practice

1. Expand $(x - 2y)^4$.
2. Find the coefficient of x^7 in $(3x + 2)^{10}$.
3. Find the constant term in $(x^3 + x^{-2})^{10}$, for $x \neq 0$.
4. Count three-person teams from six people in four ways: directly, by membership of one fixed person, by largest label, and by splitting the people into two groups of three.
5. From eight people, choose four members and designate one as chair. Count in two orders.
6. How many subsets of a nine-element set have at most four elements? Give a proof without computing five binomial coefficients.

Answers

1. $x^4 - 8x^3y + 24x^2y^2 - 32xy^3 + 16y^4$.
2. The general term is $\binom{10}{k}(3x)^{10-k}2^k$. Set $10 - k = 7$, so $k = 3$. The coefficient is $\binom{10}{3}3^72^3 = 2,099,520$.
3. The exponent is $3(10 - k) - 2k = 30 - 5k$. Set it to zero: $k = 6$. The constant term is $\binom{10}{6} = 210$.
4. Each count is 20:

$$\text{Direct: } \binom{6}{3} = 20.$$

$$\text{Include or exclude: } \binom{5}{2} + \binom{5}{3} = 10 + 10.$$

$$\text{Largest label: } \binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \binom{5}{2} = 1 + 3 + 6 + 10.$$

$$\text{Two groups: } \binom{3}{0}\binom{3}{3} + \binom{3}{1}\binom{3}{2} + \binom{3}{2}\binom{3}{1} + \binom{3}{3}\binom{3}{0} = 1 + 9 + 9 + 1.$$

5. Members first: $\binom{8}{4} \cdot 4 = 280$. Chair first: $8\binom{7}{3} = 280$, choosing three other members from the remaining seven.
6. Pair each subset with its complement. Exactly one subset in each pair has size at most four. The answer is $2^9/2 = 256$.